

We will use AWS cloud for our development and hosting. AWS provides the necessary platform support for our solution. We will also plug in open source VIRO React for creating augmented reality for airport navigation. We will design our architecture front-end using Module Federation (aka Microfrontends).

AWS Amplify

AWS Amplify is an open source development platform that empowers developers to rapidly build innovative web and mobile applications on top of the Amazon Web Services (AWS) infrastructure. It includes a wide range of open source libraries and supports drag-and-drop user interface components that developers can use as the building blocks for the desired application. AWS Amplify also has a built-in CLI (command-line interface) which can be used for app backend purposes.

It enables developers to quickly build applications with features like user authentication, offline data synchronisation, real-time updates, push notifications, etc.

AWS Amplify combines the best of AWS services, seamless integration, and an intuitive development workflow to enable developers to focus on building exceptional user experiences rather than dealing with the complexities of infrastructure management.

Key features and benefits: Here are some of its key features and benefits.

- **Seamless integration:** AWS Amplify seamlessly integrates with other AWS services, enabling developers to leverage the full potential of the AWS ecosystem. It provides easy access to services such as AWS Lambda, Amazon S3, Amazon DynamoDB, and AWS AppSync, simplifying tasks like data storage, authentication, API management, and real-time updates.
- **Rapid prototyping:** Developers can rapidly prototype and iterate on their ideas, thanks to its robust CLI and

intuitive development workflow. Using unified CLI development tasks such as creating backend resources, managing deployments and configuring continuous integration and delivery (CI/CD) pipelines can be automated.

- **Scalability and security:** Leveraging the power of AWS infrastructure, Amplify ensures scalability and security are built-in from the ground up. Applications can effortlessly scale up to handle millions of users without worrying about infrastructure provisioning. Additionally, Amplify incorporates security best practices, including encryption, user authentication, and authorisation, to protect sensitive data and ensure compliance.
- **Front-end framework support:** Amplify supports popular front-end frameworks such as React, Angular, and Vue.js, enabling developers to use their preferred tools and frameworks while benefiting from

Amplify's powerful back-end capabilities. It provides libraries and pre-built UI components that simplify common tasks like user authentication, data fetching, and real-time updates.

- **Mobile and web applications:** Amplify simplifies the development of mobile and web applications by providing pre-built UI components, backend services, and native SDKs. It enables developers to quickly build applications with features like user authentication, offline data synchronisation, real-time updates, and push notifications.
- **Serverless applications:** Amplify embraces the serverless paradigm, allowing developers to focus solely on writing application code without worrying about server management. It seamlessly integrates with AWS Lambda and other serverless services, enabling developers to create highly scalable and cost-effective applications.

Tool	Description
Amplify CLI	Command-line interface for creating, configuring, and managing backend resources.
Amplify Libraries	Collection of libraries for popular front-end frameworks (React, Angular, Vue.js, React Native) that provide pre-built UI components and API clients.
Amplify Console	Continuous deployment and hosting service for web applications, with Git integration and features like A/B testing and custom domain management.
Amplify DataStore	Scalable and persistent data synchronization solution for storing and syncing data between clients and the cloud, providing offline capabilities and realtime updates.
Amplify Authentication	Simplifies user authentication and authorization with support for various methods and integration with AWS Cognito.
Amplify Analytics	Solution for collecting and analyzing user behavior and application metrics, enabling tracking of user engagement and monitoring of application performance
Amplify Storage	Provides easy-to-use APIs for managing file and object storage, including integration with services like Amazon S3.
Amplify APIs	Simplifies the creation and management of APIs for backend services, with support for REST and GraphQL interfaces.
Amplify Machine Learning	Enables integration with AWS services for machine learning, including natural language processing, computer vision, and more

Figure 2: Amplify tools and what they do